

Coding & Solution

Date	4 July 2026
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	[Paste your GitHub / GitLab or hosted link here]
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib
Key Features Implemented	• Data preprocessing and cleaning • Feature engineering • Credit card approval prediction • Logistic Regression model

	• Decision Tree model • Random Forest model • XGBoost model • Model comparison and evaluation • Flask web application • Real-time prediction • IBM Watson Machine Learning deployment
Pending / Incomplete Features	No pending features. All planned functionalities have been implemented successfully.
Setup / Run Instructions	1. Install Python and required libraries. 2. Install dependencies using requirements.txt. 3. Run app.py. 4. Open the Flask application in a web browser. 5. Enter applicant details. 6. Click Predict to view the approval result.



Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project follows modular programming practices with separate modules for data preprocessing, feature engineering, model training, evaluation, and prediction. Four machine learning algorithms were implemented and compared, with the best-performing model integrated into a Flask web application. The solution is scalable, easy to maintain, and can be further extended with advanced security features and cloud-based deployment.